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1. What is the absolute maximum of a function?

A. The highest value of the function in the entire domain.

B. The highest value of the function in a specific interval.

C. The lowest value of the function in the domain.

D. The point where the derivative is 0.

Answer: A. The highest value of the function in the entire domain.

Explanation: The absolute maximum is the highest value a function attains in its domain.

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2. What is the absolute minimum of a function?

A. The lowest value of the function in a specific interval.

B. The lowest value of the function in the entire domain.

C. The point where the derivative is 0.

D. The point where the function is not defined.

Answer: B. The lowest value of the function in the entire domain.

Explanation: The absolute minimum is the smallest value a function attains in its domain.

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3. At which points do relative extrema occur?

A. Endpoints of the interval only.

B. Points where the derivative does not exist.

C. Critical points of the function.

D. Points where the function is undefined.

Answer: C. Critical points of the function.

Explanation: Relative extrema occur at critical points, where the derivative is 0 or undefined.

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4. A function $f(x)$ defined on a closed interval always has:

A. Only a relative maximum.

B. Only a relative minimum.

C. Both an absolute maximum and minimum.

D. No extrema.

Answer: C. Both an absolute maximum and minimum.

Explanation: By the **Extreme Value Theorem**, a continuous function on a closed interval has both absolute extrema.

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5. If $f'(x) = 0$ at $x = c$, then $x = c$ is a:

A. Relative maximum.

B. Relative minimum.

C. Critical point.

D. Absolute minimum.

Answer: C. Critical point.

Explanation: $f'(x) = 0$ indicates a critical point, which can be a maximum, minimum, or neither.

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6. How do you find the absolute extrema of a continuous function on a closed interval?

A. Check the critical points only.

B. Evaluate the endpoints only.

C. Check both critical points and endpoints.

D. Differentiate twice.

Answer: C. Check both critical points and endpoints.

Explanation: Absolute extrema occur either at critical points or endpoints.

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7. What is the condition for a function to have absolute extrema on a finite interval?

A. The function must be increasing.

B. The function must be differentiable.

C. The function must be continuous.

D. The function must be odd.

Answer: C. The function must be continuous.

Explanation: The **Extreme Value Theorem** requires continuity on a closed interval.

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8. A function $f(x)$ is given by $f(x) = x^3$. What is the absolute extrema on $[-2, 2]$?

A. Absolute maximum = 8, Absolute minimum = -8

B. Absolute maximum = 2, Absolute minimum = -2

C. Absolute maximum = 4, Absolute minimum = -4

D. None of the above

Answer: A. Absolute maximum = 8, Absolute minimum = -8

Explanation: Evaluate $f(x)$ at $x = -2, 0, 2$: $f(-2) = -8, f(2) = 8$.

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9. If $f'(x) = 0$ for all x, what can be said about $f(x)$?

A. It is increasing.

B. It is decreasing.

C. It is constant.

D. It has no extrema.

Answer: C. It is constant.

Explanation: If the derivative is 0, the function does not change and remains constant.

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10. What is the first step in solving applied maximum and minimum problems?

A. Differentiate the function.

B. Write the equation in standard form.

C. Identify the quantity to be maximized or minimized.

D. Solve the derivative equation.

Answer: C. Identify the quantity to be maximized or minimized.

Explanation: Defining the problem is the first step.

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11. What is the geometric interpretation of a relative maximum?

A. The function is the highest in the entire domain.

B. The function is the highest in its neighborhood.

C. The derivative is always positive.

D. The derivative is undefined.

Answer: B. The function is the highest in its neighborhood.

Explanation: A relative maximum is a peak point in a small interval.

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12. What happens if a function has a discontinuity in its domain?

A. It has no extrema.

B. It may not have absolute extrema.

C. It always has a maximum.

D. It always has a minimum.

Answer: B. It may not have absolute extrema.

Explanation: Discontinuity can prevent the function from meeting the conditions for extrema.

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13. What is the critical point of $f(x) = x^2 - 4x + 4$?

A. $x = 2$

B. $x = -2$

C. $x = 0$

D. $x = 4$

Answer: A. $x = 2$

Explanation: Solve $f'(x) = 2x - 4 = 0$.

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14. If a function is decreasing on an interval, then:

A. $f'(x) > 0$

B. $f'(x) < 0$

C. $f'(x) = 0$

D. $f'(x) \neq 0$

Answer: B. $f'(x) < 0$

Explanation: A negative derivative indicates the function is decreasing.

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15. What is the endpoint evaluation for $f(x) = -x^2 + 4x$ on $[0, 3]$?

A. 0 and 9

B. 4 and 3

C. 3 and 6

D. 0 and 6

Answer: D. 0 and 6

Explanation: Evaluate $f(0) = 0$ and $f(3) = 6$.

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16. In problems involving non-finite intervals, absolute extrema can occur:

A. At critical points only.

B. At infinity.

C. At endpoints.

D. At asymptotes.

Answer: B. At infinity.

Explanation: The behavior at infinity must be analyzed for non-finite intervals.

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17. A function $f(x)$ has an absolute minimum if:

A. It is unbounded below.

B. It has a lowest value in the domain.

C. Its derivative is 0.

D. It is increasing.

Answer: B. It has a lowest value in the domain.

Explanation: The absolute minimum is the lowest value the function attains.

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18. For $f(x) = 1/x$, does it have an absolute minimum on $(0, \infty)$?

A. Yes

B. No

Answer: B. No

Explanation: The function decreases without bound as $x \to 0^+$.

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1. Newton's method is used for solving:

A. Linear equations.

B. Systems of linear equations.

C. $f(x) = 0$.

D. Differential equations.

Answer: C. $f(x) = 0$.

Explanation: Newton's method approximates the roots of a function $f(x)$.

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2. What is the formula used in Newton's method?

A. $x_{n+1} = x_n - \frac{f'(x_n)}{f(x_n)}$

B. $x_{n+1} = x_n + f(x_n)f'(x_n)$

C. $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$

D. $x_{n+1} = x_n - f(x_n)f'(x_n)$

Answer: C. $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$.

Explanation: This iterative formula updates the approximation for the root.

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3. Newton's method works best when:

A. The initial guess is close to the root.

B. The derivative of $f(x)$ is zero.

C. The function is not continuous.

D. The function is linear.

Answer: A. The initial guess is close to the root.

Explanation: Convergence of Newton's method depends on a good initial guess.

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4. A difficulty in Newton's method arises when:

A. $f'(x) \neq 0$.

B. $f'(x) = 0$.

C. $f(x) = 0$.

D. $f(x)$ is continuous.

Answer: B. $f'(x) = 0$.

Explanation: When $f'(x) = 0$, the formula becomes undefined.

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5. Newton's method fails to converge if:

A. The derivative is too large.

B. The initial guess is very far from the root.

C. The function has a single root.

D. The function is quadratic.

Answer: B. The initial guess is very far from the root.

Explanation: Poor initial guesses can lead to divergence.

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6. Rolle's Theorem applies if:

A. $f(a) = f(b)$ and $f(x)$ is differentiable on (a, b).

B. $f(a) \neq f(b)$ and $f(x)$ is continuous on $[a, b]$.

C. $f(x)$ is differentiable everywhere.

D. $f(x)$ is monotonic.

Answer: A. $f(a) = f(b)$ and $f(x)$ is differentiable on (a, b).

Explanation: Rolle's theorem requires $f(x)$ to be continuous and differentiable, with equal values at the endpoints.

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7. According to Rolle's Theorem, there exists a point $c \in (a, b)$ such that:

A. $f'(c) = f(c)$.

B. $f'(c) = 0$.

C. $f(c) = 0$.

D. $f'(c) = 1$.

Answer: B. $f'(c) = 0$.

Explanation: Rolle's theorem guarantees a stationary point where $f'(c) = 0$.

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8. Which of the following is NOT a condition for Rolle's Theorem?

A. $f(x)$ is continuous on $[a, b]$.

B. $f(x)$ is differentiable on (a, b).

C. $f(a) = f(b)$.

D. $f(x)$ is strictly increasing.

Answer: D. $f(x)$ is strictly increasing.

Explanation: Rolle's theorem does not require the function to be increasing or decreasing.

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9. Which statement is true for the Mean Value Theorem (MVT)?

A. $f'(c) = 0$.

B. $f'(c) = \frac{f(a)-f(b)}{a-b}$.

C. $f'(c) = \frac{f(b)-f(a)}{b-a}$.

D. $f'(c) = \infty$.

Answer: C. $f'(c) = \frac{f(b)-f(a)}{b-a}$.

Explanation: The MVT states that there is at least one point c where the instantaneous rate of change equals the average rate of change.

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10. The Mean Value Theorem applies if:

A. $f(x)$ is differentiable on $[a, b]$.

B. $f(x)$ is continuous on $[a, b]$ and differentiable on (a, b).

C. $f(x)$ is strictly increasing.

D. $f(x)$ has a critical point.

Answer: B. $f(x)$ is continuous on $[a, b]$ and differentiable on (a, b).

Explanation: Continuity and differentiability are necessary for MVT.

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11. The Mean Value Theorem guarantees the existence of:

A. An absolute maximum.

B. An absolute minimum.

C. A point c where $f'(c) = \frac{f(b)-f(a)}{b-a}$.

D. A point c where $f(c) = 0$.

Answer: C. A point c where $f'(c) = \frac{f(b)-f(a)}{b-a}$.

Explanation: MVT guarantees such a point c.

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12. If $f(x) = x^2$, what does Rolle's Theorem say about $f(x)$ on $[-1, 1]$?

A. $f'(c) = 0$ at $c = 0$.

B. $f'(c) = 0$ at $c = 1$.

C. $f'(c) = 0$ at $c = -1$.

D. No such point exists.

Answer: A. $f'(c) = 0$ at $c = 0$.

Explanation: Since $f(-1) = f(1) = 1$, Rolle's theorem applies, and $f'(c) = 0$ at $c = 0$.

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13. Which of the following is a practical application of the Mean Value Theorem?

A. Estimating the roots of a function.

B. Predicting the behavior of continuous functions.

C. Proving the existence of extrema.

D. Determining speed in physics.

Answer: D. Determining speed in physics.

Explanation: The MVT is often used to find average and instantaneous rates, such as speed.

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14. Newton's method may fail if:

A. The function has a horizontal tangent.

B. The initial guess is close to the root.

C. The function is linear.

D. The function is strictly increasing.

Answer: A. The function has a horizontal tangent.

Explanation: A horizontal tangent ($f'(x) = 0$) makes the method undefined.

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15. Rolle's theorem requires:

A. $f(a) = f(b)$.

B. $f(a) \neq f(b)$.

C. $f'(a) = f'(b)$.

D. $f(a) + f(b) = 0$.

Answer: A. $f(a) = f(b)$.

Explanation: Equal endpoint values are essential for Rolle's theorem.

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1. Integration is also known as:

A. Differentiation.

B. Anti-differentiation.

C. Optimization.

D. Approximation.

Answer: B. Anti-differentiation.

Explanation: Integration reverses the process of differentiation.

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2. What is the main purpose of integration in calculus?

A. Finding derivatives.

B. Finding slopes.

C. Finding areas under curves.

D. Finding tangents.

Answer: C. Finding areas under curves.

Explanation: Integration calculates the area under a curve or between curves.

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3. The integral of $f'(x) = x^n$ is:

A. $\frac{x^{n+1}}{n+1} + C$.

B. x^{n+1}.

C. nx^{n-1}.

D. $\frac{n+1}{x^n} + C$.

Answer: A. $\frac{x^{n+1}}{n+1} + C$.

Explanation: The power rule for integration is used.

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4. The constant C in an indefinite integral is called:

A. The constant of differentiation.

B. The integration multiplier.

C. The constant of integration.

D. The indefinite coefficient.

Answer: C. The constant of integration.

Explanation: It accounts for all possible vertical shifts of the antiderivative.

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5. Which of the following represents the area problem in calculus?

A. Finding the tangent to a curve.

B. Calculating the instantaneous rate of change.

C. Determining the area under a curve.

D. Optimizing functions.

Answer: C. Determining the area under a curve.

Explanation: The area problem involves finding the region between a curve and the x-axis.

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6. The integral of $\cos(x)$ is:

A. $\sin(x) + C$.

B. $-\sin(x) + C$.

C. $\cos(x) + C$.

D. $-\cos(x) + C$.

Answer: A. $\sin(x) + C$.

Explanation: The derivative of $\sin(x)$ is $\cos(x)$, so its integral is $\sin(x) + C$.

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7. Which of the following is an indefinite integral?

A. $\int_0^1 x^2 dx.$

B. $\int x^2 dx.$

C. $\int_0^a x dx.$

D. $\int_{-a}^a x dx.$

Answer: B. $\int x^2 dx.$

Explanation: Indefinite integrals do not have specified limits.

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8. What is the integral of e^x?

A. $\ln(x) + C$.

B. $e^{x+1} + C$.

C. $e^x + C$.

D. $xe^{x-1} + C$.

Answer: C. $e^x + C$.

Explanation: The derivative and integral of e^x are the same.

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9. Which property of indefinite integrals is correct?

A. $\int [f(x) + g(x)]dx = \int f(x)dx + \int g(x)dx.$

B. $\int [f(x) - g(x)]dx = \int f(x)dx - \int g(x)dx.$

C. Both A and B.

D. Neither A nor B.

Answer: C. Both A and B.

Explanation: The linearity property applies to indefinite integrals.

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10. The integral $\int \frac{1}{x} dx$ is:

A. $\ln |x| + C$.

B. $\ln(x) + C$.

C. $x^{-1} + C$.

D. $\frac{1}{x^2} + C$.

Answer: A. $\ln |x| + C$.

Explanation: The integral of $\frac{1}{x}$ requires the natural logarithm of the absolute value of x.

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11. If $F(x)$ is the antiderivative of $f(x)$, which of the following is true?

A. $F'(x) = f'(x)$.

B. $F(x) = f(x) + C$.

C. $F'(x) = f(x)$.

D. $F(x) = f'(x)$.

Answer: C. $F'(x) = f(x)$.

Explanation: Differentiation of $F(x)$ gives the original function $f(x)$.

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12. What is the indefinite integral of $\sin(x)$?

A. $-\cos(x) + C$.

B. $\cos(x) + C$.

C. $\sin(x) + C$.

D. $-\sin(x) + C$.

Answer: A. $-\cos(x) + C$.

Explanation: The derivative of $-\cos(x)$ is $\sin(x)$.

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13. Which formula is used for the integration of a sum of functions?

A. $\int [f(x) + g(x)]dx = \int f(x)dx + \int g(x)dx.$

B. $\int [f(x) + g(x)]dx = f(x) + \int g(x)dx.$

C. $\int [f(x) + g(x)]dx = \int f(x)dx - \int g(x)dx.$

D. $\int [f(x) + g(x)]dx = 0.$

Answer: A. $\int [f(x) + g(x)]dx = \int f(x)dx + \int g(x)dx.$

Explanation: This is the linearity property of integrals.

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14. Which of the following represents the property of an indefinite integral?

A. $\int [cf(x)]dx = c\int f(x)dx.$

B. $\int [cf(x)]dx = \frac{1}{c}\int f(x)dx.$

C. $\int [cf(x)]dx = f(x) + c.$

D. $\int [cf(x)]dx = \int f(x)dx.$

Answer: A. $\int [cf(x)]dx = c\int f(x)dx.$

Explanation: The constant multiplier property applies to integrals.

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15. The integral $\int 0dx$ equals:

A. 0.

B. C.

C. x.

D. $0 + C$.

Answer: B. C.

Explanation: The integral of zero is a constant.

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17. If $\int f(x)dx = g(x) + C$, then $\frac{d}{dx}[\int f(x)dx] = ?$:

A. $f'(x)$.

B. $g'(x)$.

C. $g(x)$.

D. $f(x)$.

Answer: D. $f(x)$.

Explanation: Differentiating the integral restores the original function.

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1. What does the Greek symbol Σ represent in mathematics?

A. Product of terms.

B. Summation of terms.

C. Division of terms.

D. Subtraction of terms.

Answer: B. Summation of terms.

Explanation: Σ is used to represent the sum of a sequence of terms.

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2. In Σ notation, the symbol i is typically referred to as:

A. The upper bound.

B. The lower bound.

C. The index of summation.

D. The sum constant.

Answer: C. The index of summation.

Explanation: i is the variable that iterates through the sequence.

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3. What does $\sum_{i=1}^{n} a_i$ mean?

A. Multiply all terms a_i from $i = 1$ to n.

B. Add all terms a_i from $i = 1$ to n.

C. Subtract all terms a_i from $i = 1$ to n.

D. Divide all terms a_i from $i = 1$ to n.

Answer: B. Add all terms a_i from $i = 1$ to n.

Explanation: Σ adds the sequence of terms defined by a_i.

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3. What does $\sum_{i=1}^{n} a_i$ mean?

A. Multiply all terms a_i from $i = 1$ to n.

B. Add all terms a_i from $i = 1$ to n.

C. Subtract all terms a_i from $i = 1$ to n.

D. Divide all terms a_i from $i = 1$ to n.

Answer: B. Add all terms a_i from $i = 1$ to n.

Explanation: Σ adds the sequence of terms defined by a_i.

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5. In $\sum_{i=1}^{n} c$, where c is a constant, the result is:

A. c.

B. nc.

C. c^n.

D. $\frac{c}{n}$.

Answer: B. nc.

Explanation: Adding c, n times results in nc.

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6. What is the simplified value of $\sum_{i=1}^{3}(2i+1)$?

A. 12.

B. 15.

C. 18.

D. 21.

Answer: B. 15.

Explanation: Compute $(2(1)+1)+(2(2)+1)+(2(3)+1)=3+5+7=15$.

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7. Which of the following is a correct property of Σ notation?

A. $\sum_{i=1}^{n}[a_i + b_i] = \sum_{i=1}^{n} a_i - \sum_{i=1}^{n} b_i.$

B. $\sum_{i=1}^{n}[a_i + b_i] = \sum_{i=1}^{n} a_i + \sum_{i=1}^{n} b_i.$

C. $\sum_{i=1}^{n}[a_i + b_i] = \sum_{i=1}^{n} a_i \times \sum_{i=1}^{n} b_i.$

D. $\sum_{i=1}^{n}[a_i + b_i] = \sum_{i=1}^{n} b_i.$

Answer: B. $\sum_{i=1}^{n}[a_i + b_i] = \sum_{i=1}^{n} a_i + \sum_{i=1}^{n} b_i.$

Explanation: This is the property of linearity of summation.

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8. What does the expression $\sum_{k=3}^{5} k^2$ evaluate to?

A. 50.

B. 55.

C. 59.

D. 65.

Answer: C. 59.

Explanation: Calculate $3^2 + 4^2 + 5^2 = 9 + 16 + 25 = 59$.

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9. If $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$, what is the value of $\sum_{i=1}^{10} i$?

A. 45.

B. 55.

C. 60.

D. 65.

Answer: B. 55.

Explanation: Substitute $n = 10$ into $\frac{n(n+1)}{2}$: $\frac{10(10+1)}{2} = 55$.

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10. For $\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$, what is $\sum_{i=1}^{4} i^2$?

A. 30.

B. 35.

C. 40.

D. 45.

Answer: A. 30.

Explanation: Substitute $n = 4$ into $\frac{n(n+1)(2n+1)}{6}$: $\frac{4(4+1)(2(4)+1)}{6} = 30$.

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11. If $\sum_{i=1}^{n} c$ equals nc, what is $\sum_{i=1}^{5} 3$?

A. 3.

B. 9.

C. 12.

D. 15.

Answer: D. 15.

Explanation: Add 3, 5 times: $5 \times 3 = 15$.

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12. Which of the following is true about Σ notation?

A. It is used to represent subtraction.

B. It is used to write lengthy products compactly.

C. It is used to write lengthy sums compactly.

D. It cannot represent arithmetic series.

Answer: C. It is used to write lengthy sums compactly.

Explanation: Σ notation simplifies representing long summations.

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13. The value of $\sum_{i=1}^{n}[3i-1]$ for $n=3$ is:

A. 15.

B. 20.

C. 25.

D. 30.

Answer: B. 20.

Explanation: Substitute $i=1,2,3$: $(3(1)-1)+(3(2)-1)+(3(3)-1)=2+5+8=20$.

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14. The expression $\sum_{k=1}^{n} 2k$ simplifies to:

A. $n(n+1)$.

B. $n(n-1)$.

C. $2n(n+1)$.

D. n^2+1.

Answer: C. $2n(n+1)$.

Explanation: Use the property $\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$ and multiply by 2.

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15. What is $\sum_{i=1}^{4} 2^i$?

A. 30.

B. 31.

C. 32.

D. 33.

Answer: B. 31.

Explanation: Compute $2^1 + 2^2 + 2^3 + 2^4 = 2 + 4 + 8 + 16 = 31$.

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1. What does the Definite Integral of a function $f(x)$ over an interval $[a, b]$ represent?

A. The derivative of $f(x)$ at $x = a$.

B. The total area under the curve $f(x)$ between $x = a$ and $x = b$.

C. The slope of the tangent line at $x = b$.

D. The maximum value of $f(x)$ on $[a, b]$.

Answer: B. The total area under the curve $f(x)$ between $x = a$ and $x = b$.

Explanation: The Definite Integral calculates the net area under the curve from $x = a$ to $x = b$.

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2. If $f(x) \geq 0$ on $[a, b]$, then the Definite Integral $\int_a^b f(x)\,dx$:

A. Will always be negative.

B. Will always be zero.

C. Will always be positive or zero.

D. Can be either positive or negative.

Answer: C. Will always be positive or zero.

Explanation: Since $f(x) \geq 0$, the area under the curve is non-negative.

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3. If $f(x)$ has both positive and negative values on $[a, b]$, the Definite Integral $\int_a^b f(x)\,dx$:

A. Represents the net area, considering signs of $f(x)$.

B. Represents only the positive area.

C. Ignores negative values of $f(x)$.

D. Always equals zero.

Answer: A. Represents the net area, considering signs of $f(x)$.

Explanation: The Definite Integral sums positive areas as positive and negative areas as negative.

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4. What happens to the Definite Integral $\int_a^b f(x)\,dx$ if $f(x)$ is discontinuous at one point in $[a, b]$?

A. It is undefined.

B. It can still be calculated if $f(x)$ is piecewise continuous.

C. It is always zero.

D. It depends on the type of discontinuity.

Answer: B. It can still be calculated if $f(x)$ is piecewise continuous.

Explanation: The Definite Integral can handle removable or jump discontinuities as long as the function is piecewise continuous.

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5. Which property of Definite Integrals is correct?

A. $\int_a^b [f(x) + g(x)]\,dx = \int_a^b f(x)\,dx - \int_a^b g(x)\,dx.$

B. $\int_a^b c\,dx = c(b-a)$, where c is a constant.

C. $\int_a^b f(x)\,dx = \int_b^a f(x)\,dx.$

D. $\int_a^b f(x)\,dx = -\int_a^b f(x)\,dx.$

Answer: B. $\int_a^b c\,dx = c(b-a)$, where c is a constant.

Explanation: This is a standard property of Definite Integrals for a constant function.

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6. If $\int_a^b f(x)\,dx = 10$, what is the value of $\int_b^a f(x)\,dx$?

A. 10

B. -10

C. 0

D. Undefined

Answer: B. -10

Explanation: Reversing the limits of integration negates the value of the integral.

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7. For $f(x) \geq g(x)$ on $[a, b]$, which of the following is true?

A. $\int_a^b f(x)\,dx < \int_a^b g(x)\,dx.$

B. $\int_a^b f(x)\,dx = \int_a^b g(x)\,dx.$

C. $\int_a^b f(x)\,dx > \int_a^b g(x)\,dx.$

D. $\int_a^b f(x)\,dx$ is always zero.

Answer: C. $\int_a^b f(x)\,dx > \int_a^b g(x)\,dx.$

Explanation: If $f(x) \geq g(x)$, the area under $f(x)$ is greater than or equal to the area under $g(x)$.

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8. Which of the following inequalities involving Definite Integrals is correct?

A. $\int_a^b f(x)\,dx \leq 0$ if $f(x) \geq 0$.

B. $\int_a^b f(x)\,dx \geq \int_a^b g(x)\,dx$ if $f(x) \geq g(x)$ on $[a, b]$.

C. $\int_a^b f(x)\,dx > \int_a^b g(x)\,dx$ for all $f(x)$ and $g(x)$.

D. $\int_a^b f(x)\,dx = \int_a^b g(x)\,dx$ for $f(x) \neq g(x)$.

Answer: B. $\int_a^b f(x)\,dx \geq \int_a^b g(x)\,dx$ if $f(x) \geq g(x)$ on $[a, b]$.

Explanation: This inequality holds if $f(x) \geq g(x)$ throughout the interval.

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9. What is the value of $\int_a^a f(x)\,dx$?

A. 0

B. $f(a)$

C. a^2

D. Undefined

Answer: A. 0

Explanation: The Definite Integral over an interval where $a = b$ is always zero.

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10. If $f(x) = 3x^2$, what is $\int_1^2 f(x)\,dx$?

A. 7

B. 10

C. 12

D. 15

Answer: C. 12

Explanation: Compute $\int_1^2 3x^2\,dx = [x^3]_1^2 = 2^3 - 1^3 = 8 - 1 = 12.$

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1. What does the First Fundamental Theorem of Calculus state?

A. Every continuous function has a derivative.

B. Every differentiable function has an integral.

C. The definite integral of a function can be computed using its antiderivative.

D. The derivative of a function is equal to its integral.

Answer: C. The definite integral of a function can be computed using its antiderivative.

Explanation: The First Fundamental Theorem of Calculus links differentiation and integration.

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Which of the following is NOT a measure of central tendency?

- **A) Mean**
- **B) Median**
- **C) Mode**
- **D) Variance**

Answer: D) Variance

Explanation: Variance is a measure of the spread or dispersion of data, not a measure of central tendency.

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2. If $F(x)$ is an antiderivative of $f(x)$, then the definite integral $\int_a^b f(x)\,dx$ is equal to:

A. $F(a) - F(b)$

B. $F(b) - F(a)$

C. $F'(b) - F'(a)$

D. $F(a) + F(b)$

Answer: B. $F(b) - F(a)$

Explanation: The First Fundamental Theorem of Calculus states that $\int_a^b f(x)\,dx = F(b) - F(a)$.

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4. What does the Mean Value Theorem for Integrals state?

A. The integral of a function is equal to the sum of its derivatives.

B. There exists a point c in $[a, b]$ such that $f(c) = \frac{1}{b-a} \int_a^b f(x)\,dx$.

C. The value of a function is constant on $[a, b]$.

D. The function is equal to its antiderivative at c.

Answer: B. There exists a point c in $[a, b]$ such that $f(c) = \frac{1}{b-a} \int_a^b f(x)\,dx$.

Explanation: The Mean Value Theorem for Integrals guarantees the existence of such a point c.

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5. The average value of a function $f(x)$ on an interval $[a, b]$ is given by:

A. $\frac{\int_a^b f(x)\,dx}{b-a}$

B. $\int_a^b f(x)\,dx \cdot (b-a)$

C. $\frac{b-a}{\int_a^b f(x)\,dx}$

D. $f'(b) - f'(a)$

Answer: A. $\frac{\int_a^b f(x)\,dx}{b-a}$

Explanation: The average value is the total value divided by the interval length.

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6. If $f(x)$ is continuous on $[a, b]$, then according to the First Fundamental Theorem of Calculus:

A. $\frac{d}{dx} \int_a^x f(t)\, dt = f(x)$.
B. $\int_a^b f(x)\, dx = 0$.
C. $f(a) = f(b)$.
D. $f(x)$ is differentiable on $[a, b]$.
Answer: A. $\frac{d}{dx} \int_a^x f(t)\, dt = f(x)$.
Explanation: The theorem states that the derivative of the integral function returns the original function.

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7. For a constant function $f(x) = c$, the average value on $[a, b]$ is:

A. 0

B. $c(b - a)$

C. c

D. $\frac{b+a}{2}$

Answer: C. c

Explanation: The average value of a constant function is the constant itself.

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9. If $F'(x) = f(x)$, then $\int_a^b f(x)\,dx$:

A. Depends on $F'(x)$.

B. Is independent of the limits a and b.

C. Depends only on $F(b)$ and $F(a)$.

D. Is undefined if $f(x)$ is not differentiable.

Answer: C. Depends only on $F(b)$ and $F(a)$.

Explanation: The integral depends only on the antiderivative evaluated at the bounds.

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1. What is the method of slicing used for in calculus?

A. To approximate areas under curves.

B. To find volumes of three-dimensional solids by integrating cross-sectional areas.

C. To find the derivative of a function.

D. To calculate the tangent of a curve.

Answer: B. To find volumes of three-dimensional solids by integrating cross-sectional areas.

Explanation: The method of slicing involves cutting the solid into thin slices and integrating their areas to find the volume.

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2. Which of the following represents the formula to find the volume of a solid using the disk method when the slices are perpendicular to the x-axis?

A. $V = \int_a^b \pi \left[R(x)^2 - r(x)^2\right] dx$

B. $V = \int_a^b \pi \left[R(y)^2 - r(y)^2\right] dy$

C. $V = \int_a^b \pi R(x)^2 dx$

D. $V = \int_a^b \pi r(x)^2 dx$

Answer: C. $V = \int_a^b \pi R(x)^2 dx$

Explanation: This formula is used for finding volumes by disks when the cross-sections are perpendicular to the x-axis.

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3. When using the washer method for solids of revolution, which formula is applied when the cross-sections are perpendicular to the x-axis?

A. $V = \int_a^b \pi \left[R(x)^2 - r(x)^2\right] dx$
B. $V = \int_a^b \pi R(y)^2 dy$
C. $V = \int_a^b \pi r(x)^2 dx$
D. $V = \int_a^b \pi \left[R(x)^2 + r(x)^2\right] dx$
Answer: A. $V = \int_a^b \pi \left[R(x)^2 - r(x)^2\right] dx$

Explanation: The washer method uses the outer radius $R(x)$ and inner radius $r(x)$ to subtract the inner area from the outer area, which is then integrated.

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4. In the disk method, if the slices are taken perpendicular to the y-axis, the formula for volume is:

A. $V = \int_a^b \pi \left[R(x)^2\right] dx$
B. $V = \int_a^b \pi \left[R(y)^2\right] dy$
C. $V = \int_a^b \pi \left[r(x)^2\right] dx$
D. $V = \int_a^b \pi \left[r(y)^2\right] dy$
Answer: B. $V = \int_a^b \pi \left[R(y)^2\right] dy$

Explanation: The disk method with slices perpendicular to the y-axis integrates the square of the radius function with respect to y.

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5. When calculating volumes of solids of revolution, which method is used when the region is revolved about the x-axis?

A. The method of cross-sections.

B. The disk or washer method.

C. The method of slicing.

D. The trapezoidal method.

Answer: B. The disk or washer method.

Explanation: The disk or washer method is commonly used to calculate volumes of solids formed by revolving a region around an axis.

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6. What is the volume of a cylinder with radius r and height h obtained using the disk method?

A. $V = \pi r^2 h$

B. $V = \int_0^h \pi r^2 dx$

C. $V = \int_0^r \pi h^2 dy$

D. $V = \pi h^3$

Answer: A. $V = \pi r^2 h$

Explanation: The volume of a cylinder is directly given by $\pi r^2 h$, which is an example of using the disk method.

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7. In the washer method, the volume of a solid formed by revolving a region around the y-axis is given by:

A. $V = \int_a^b \pi \left[R(y)^2 - r(y)^2\right] dy$
B. $V = \int_a^b \pi \left[R(x)^2 - r(x)^2\right] dx$
C. $V = \int_a^b \pi R(x)^2 dx$
D. $V = \int_a^b \pi R(y)^2 dy$
Answer: A. $V = \int_a^b \pi \left[R(y)^2 - r(y)^2\right] dy$

Explanation: This formula is used for finding the volume of a solid when revolving around the y-axis using the washer method.

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8. Which of the following is an appropriate method for finding the volume of a solid formed by rotating a region around the x-axis?

A. Cross-sections perpendicular to the x-axis.

B. Cross-sections perpendicular to the y-axis.

C. The disk method for x-axis revolution.

D. The trapezoidal rule.

Answer: C. The disk method for x-axis revolution.

Explanation: When a region is rotated around the x-axis, the disk method is commonly used to compute the volume.

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9. The volume of a solid generated by rotating a region around the x-axis can be found using:

A. $V = \int_a^b \pi f(x)^2 dx$
B. $V = \int_a^b \pi \left[f(x) - g(x)\right]^2 dx$
C. $V = \int_a^b \pi f(x) dx$
D. $V = \int_a^b \pi \left[g(x) - f(x)\right]^2 dx$
Answer: A. $V = \int_a^b \pi f(x)^2 dx$

Explanation: This formula is used for finding volumes of solids formed by rotating a region around the x-axis, with the function $f(x)$ representing the radius of the disk.

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10. To calculate the volume of a solid by cross-sections perpendicular to the x-axis, we use:

A. The method of slicing with washers.

B. The method of slicing with disks.

C. Integration of cross-sectional areas with respect to x.

D. The method of cylindrical shells.

Answer: C. Integration of cross-sectional areas with respect to x.

Explanation: When calculating volumes using cross-sections perpendicular to the x-axis, the areas of the slices are integrated with respect to x.

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1. What is the formula for work done by a constant force F over a displacement d?

A. $W = F \cdot d$

B. $W = F \cdot \frac{1}{2}d^2$

C. $W = \int_0^d F\,dx$

D. $W = F \cdot \sin(\theta) \cdot d$

Answer: A. $W = F \cdot d$

Explanation: Work done by a constant force is given by the product of the force and the displacement in the direction of the force.

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2. The work done by a variable force is calculated using:

A. $W = \int_0^d F\,dx$

B. $W = F \cdot d$

C. $W = \int_a^b F(x)\,dx$

D. $W = F \cdot d^2$

Answer: C. $W = \int_a^b F(x)\,dx$

Explanation: When the force is variable, work is calculated by integrating the force function over the displacement.

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3. Which of the following describes fluid pressure at a point in a fluid at rest?

A. The force per unit area exerted on the fluid.

B. The total force exerted by the fluid on an object.

C. The velocity of the fluid at that point.

D. The temperature of the fluid at that point.

Answer: A. The force per unit area exerted on the fluid.

Explanation: Fluid pressure is the force exerted by the fluid per unit area at a given point in the fluid.

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4. Pascal's Principle states that pressure applied to an enclosed fluid is:

A. Transmitted equally in all directions.

B. Only transmitted downward.

C. Only transmitted horizontally.

D. Only transmitted to the surface of the fluid.

Answer: A. Transmitted equally in all directions.

Explanation: According to Pascal's Principle, any change in pressure applied to an enclosed fluid is transmitted undiminished in all directions throughout the fluid.

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5. In calculating the work done by a variable force, the integral $\int_a^b F(x)\,dx$ represents:

A. The force over the distance.

B. The total work done as the force varies.

C. The change in force over the displacement.

D. The average force over the interval.

Answer: B. The total work done as the force varies.

Explanation: The integral of the force function over a displacement gives the total work done when the force is not constant.

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6. The work done by a constant force F when applied at an angle θ to the direction of displacement is given by:

A. $W = F \cdot d$

B. $W = F \cdot \cos(\theta) \cdot d$

C. $W = F \cdot d^2$

D. $W = F \cdot \sin(\theta) \cdot d$

Answer: B. $W = F \cdot \cos(\theta) \cdot d$

Explanation: The work done is calculated by multiplying the force, the displacement, and the cosine of the angle between the force and the displacement direction.

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7. If a force is applied on an object in the direction of motion, the work done is:

A. Zero.

B. Positive.

C. Negative.

D. Undefined.

Answer: B. Positive.

Explanation: When the force is in the same direction as the motion, the work done is positive.

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8. Fluid pressure at a given depth in a fluid depends on:

A. The density of the fluid and the acceleration due to gravity.

B. The temperature of the fluid and the fluid's color.

C. The velocity of the fluid and the area of contact.

D. The height of the fluid and the object's shape.

Answer: A. The density of the fluid and the acceleration due to gravity.

Explanation: Fluid pressure increases with depth and is given by $P = \rho g h$, where ρ is the fluid density, g is the acceleration due to gravity, and h is the depth.

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9. Work done in a variable force situation can be found by calculating the integral of force over which variable?

A. Force.

B. Distance.

C. Time.

D. Energy.

Answer: B. Distance.

Explanation: Work is the integral of the force over distance when the force is variable.

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10. In the context of Pascal's Principle, the pressure in a hydraulic press is transmitted:

A. Only to the top of the fluid.

B. Unequally to different points in the fluid.

C. Equally in all directions.

D. Only vertically.

Answer: C. Equally in all directions.

Explanation: Pascal's Principle states that pressure applied at any point in an enclosed fluid is transmitted equally in all directions.

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1. What is an improper integral?

A. An integral with a constant limit.

B. An integral involving a function with infinite values or an infinite interval.

C. An integral that is always convergent.

D. An integral that is always divergent.

Answer: B. An integral involving a function with infinite values or an infinite interval.

Explanation: Improper integrals involve either infinite limits or integrands that become infinite at certain points.

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2. Which of the following describes an improper integral over an infinite interval?

A. $\int_0^\infty f(x)\,dx$
B. $\int_{-1}^1 f(x)\,dx$
C. $\int_0^1 f(x)\,dx$
D. $\int_{-5}^5 f(x)\,dx$

Answer: A. $\int_0^\infty f(x)\,dx$

Explanation: An improper integral over an infinite interval involves an infinite limit, such as $\int_0^\infty f(x)\,dx$.

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3. The improper integral $\int_1^\infty \frac{1}{x^2}\,dx$ is:

A. Convergent.

B. Divergent.

C. Undefined.

D. Zero.

Answer: A. Convergent.

Explanation: This improper integral converges because the function $\frac{1}{x^2}$ decays fast enough as $x \to \infty$.

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If the data set is skewed to the right, which of the following is the correct order of central tendencies?

- **A) Mode > Median > Mean**
- **B) Mean > Median > Mode**
- **C) Mean > Mode > Median**
- **D) Mode > Mean > Median**

Answer: B) Mean > Median > Mode

Explanation: In right-skewed data, the mean is greater than the median, which is greater than the mode.

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4. When evaluating an improper integral of the form $\int_a^{\infty} f(x)\,dx$, if the limit does not exist, the integral is:

A. Convergent.

B. Divergent.

C. Undefined.

D. Equal to zero.

Answer: B. Divergent.

Explanation: If the limit does not exist as $x \to \infty$, the improper integral is divergent.

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5. Which of the following represents an improper integral where the integrand becomes infinite at a point?

A. $\int_0^1 \frac{1}{x}\,dx$

B. $\int_1^5 \frac{1}{x}\,dx$

C. $\int_{-1}^1 x^2\,dx$

D. $\int_2^4 x^3\,dx$

Answer: A. $\int_0^1 \frac{1}{x}\,dx$

Explanation: The integrand $\frac{1}{x}$ becomes infinite as $x \to 0$, making it an improper integral.

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6. For the integral $\int_1^\infty \frac{1}{x^p}\,dx$, the integral is convergent if:

A. $p \geq 1$

B. $p < 1$

C. $p = 1$

D. $p > 1$

Answer: D. $p > 1$

Explanation: The integral $\int_1^\infty \frac{1}{x^p}\,dx$ converges if $p > 1$.

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8. Which of the following is a technique used to evaluate improper integrals over infinite intervals?

A. Substitution method.

B. Limit approach.

C. Partial fractions.

D. Integration by parts.

Answer: B. Limit approach.

Explanation: To evaluate improper integrals over infinite intervals, the limit approach is used, where you take the limit of the integral as the variable approaches infinity.

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10. The improper integral $\int_1^\infty e^{-x}\,dx$ is:

A. Divergent.

B. Undefined.

C. Convergent.

D. Zero.

Answer: C. Convergent.

Explanation: The integral $\int_1^\infty e^{-x}\,dx$ converges because e^{-x} decays exponentially as $x \to \infty$.

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1. What is a sequence?

A. A set of unordered numbers.

B. A collection of numbers ordered by their position.

C. A single number in a series.

D. A geometric progression.

Answer: B. A collection of numbers ordered by their position.

Explanation: A sequence is a list of numbers arranged in a specific order, where each number is identified by its position in the list.

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2. Which of the following is the graph of a sequence?

A. A continuous curve.

B. A scatter plot of points.

C. A line segment connecting the points.

D. A set of unconnected points.

Answer: D. A set of unconnected points.

Explanation: The graph of a sequence is typically represented as a set of points on the Cartesian plane, where the x-coordinate is the position of the term in the sequence and the y-coordinate is the value of the term.

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3. What is the limit of a sequence?

A. The sum of all the terms.

B. The value that the terms of the sequence approach as the index increases.

C. The first term of the sequence.

D. The maximum value in the sequence.

Answer: B. The value that the terms of the sequence approach as the index increases.

Explanation: The limit of a sequence is the value that the terms get closer to as the index goes to infinity.

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4. Which of the following is an example of a recursive sequence?

A. $a_n = 2n$

B. $a_n = 5n + 3$

C. $a_n = a_{n-1} + 2$

D. $a_n = 3n - 1$

Answer: C. $a_n = a_{n-1} + 2$

Explanation: A recursive sequence is defined in terms of its previous terms, like $a_n = a_{n-1} + 2$, where each term depends on the previous one.

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5. How can you test for monotonicity in a sequence?

A. By checking if the sequence has a limit.

B. By analyzing the difference between consecutive terms.

C. By plotting the sequence on a graph.

D. By finding the sum of the terms.

Answer: B. By analyzing the difference between consecutive terms.

Explanation: To test for monotonicity, we check whether the sequence is increasing or decreasing by examining the difference between consecutive terms: if $a_{n+1} - a_n > 0$, the sequence is increasing, and if $a_{n+1} - a_n < 0$, it is decreasing.

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6. What does it mean for a sequence to be eventually monotonic?

A. It is always increasing or always decreasing.

B. It eventually becomes constant.

C. After some point, it becomes either increasing or decreasing.

D. It oscillates between values.

Answer: C. After some point, it becomes either increasing or decreasing.

Explanation: A sequence is eventually monotonic if, beyond some index, it becomes either strictly increasing or strictly decreasing.

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7. What is true about the convergence of a monotonic sequence?

A. It must always converge.

B. It may converge or diverge.

C. It always diverges.

D. It oscillates indefinitely.

Answer: B. It may converge or diverge.

Explanation: A monotonic sequence can either converge to a limit or diverge depending on whether it is bounded. A bounded monotonic sequence converges, while an unbounded one diverges.

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8. Which of the following statements is true about a convergent sequence?

A. It has a limit.

B. It oscillates between values.

C. It increases without bound.

D. It has no limit.

Answer: A. It has a limit.

Explanation: A convergent sequence has a limit, meaning its terms approach a specific value as the index increases.

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The variance of a data set is the square of which of the following?

- **A) The mean**
- **B) The standard deviation**
- **C) The mode**
- **D) The median**

Answer: B) The standard deviation

Explanation: Variance is the square of the standard deviation, representing the average squared deviation from the mean.

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If the variance of a data set is 25, what is the standard deviation?

- A) 5
- B) 25
- C) 12
- D) 50

Answer: A) 5

Explanation: The standard deviation is the square root of the variance, so 25=5\sqrt{25} = 525=5.

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9. Which of the following is an example of a monotonically increasing sequence?

A. $a_n = (-1)^n$

B. $a_n = 3n + 1$

C. $a_n = \frac{1}{n}$

D. $a_n = n^2$

Answer: B. $a_n = 3n + 1$

Explanation: The sequence $a_n = 3n + 1$ is monotonically increasing because each term is larger than the previous one.

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10. What is the intuitive idea behind the convergence of a sequence?

A. The sequence oscillates without a limit.

B. The sequence approaches a single value as the index increases.

C. The sequence grows without bound.

D. The sequence stops at a fixed value.

Answer: B. The sequence approaches a single value as the index increases.

Explanation: Convergence means that the terms of the sequence get arbitrarily close to a single value as the index tends to infinity.

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Derivative of f for $x \geq 1$	Classification of the sequence with $f(x) = a_x$
$f'(x) > 0$	Increasing
$f'(x) < 0$	Decreasing
$f'(x) \geq 0$	Nondecreasing
$f'(x) \leq 0$	Nonincreasing

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Difference between successive terms	Classification
$a_{n+1} - a_n > 0$	Increasing
$a_{n+1} - a_n < 0$	Decreasing
$a_{n+1} - a_n \geq 0$	Nondecreasing
$a_{n+1} - a_n \leq 0$	Nonincreasing

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Ratio of successive terms	Classification
$a_{n+1} / a_n > 1$	Increasing
$a_{n+1} / a_n < 1$	Decreasing
$a_{n+1} / a_n \geq 1$	Nondecreasing
$a_{n+1} / a_n \leq 1$	Nonincreasing

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In a normal distribution, about what percentage of the data falls within one standard deviation of the mean?

- **A) 50%**
- **B) 68%**
- **C) 95%**
- **D) 99%**

Answer: B) 68%

Explanation: In a normal distribution, approximately 68% of the data falls within one standard deviation of the mean.

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1. What is an alternating series?

A. A series with terms that always have positive values.

B. A series with terms that alternate between positive and negative values.

C. A series that sums to zero.

D. A series that converges only for even terms.

Answer: B. A series with terms that alternate between positive and negative values.

Explanation: An alternating series has terms where the signs alternate between positive and negative.

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1. What is an alternating series?

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2. Which of the following is an example of an alternating series?

A. $1 + 2 + 3 + 4 + 5 + \dots$

B. $1 - 2 + 3 - 4 + 5 - \dots$

C. $1 + 3 + 5 + 7 + 9 + \dots$

D. $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots$

Answer: B. $1 - 2 + 3 - 4 + 5 - \dots$

Explanation: This series has terms that alternate between positive and negative values, which defines an alternating series.

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3. What does conditional convergence refer to?

A. A series that converges absolutely.

B. A series that converges under specific conditions.

C. A series that converges only when rearranged.

D. A series that diverges.

Answer: B. A series that converges under specific conditions.

Explanation: Conditional convergence occurs when a series converges, but it does not converge absolutely. Its convergence depends on the order of the terms.

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4. Which test is used to determine if an alternating series converges?

A. The Ratio Test

B. The Alternating Series Test (Leibniz's Test)

C. The Root Test

D. The Integral Test

Answer: B. The Alternating Series Test (Leibniz's Test)

Explanation: Leibniz's test provides a criterion for the convergence of alternating series, which involves checking if the terms decrease in absolute value and tend to zero.

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5. For an alternating series $\sum(-1)^n a_n$, which of the following is required for the series to converge according to the Alternating Series Test?

A. The sequence a_n must increase.

B. The sequence a_n must decrease.

C. The sequence a_n must oscillate.

D. The sequence a_n must be constant.

Answer: B. The sequence a_n must decrease.

Explanation: According to the Alternating Series Test, for an alternating series to converge, the terms must decrease in absolute value and approach zero.

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6. Which of the following is a condition for absolute convergence of a series?

A. The series must have a finite number of terms.

B. The series must sum to a finite number even when all terms are made positive.

C. The series must have an alternating sign.

D. The series must diverge when the terms are positive.

Answer: B. The series must sum to a finite number even when all terms are made positive.

Explanation: Absolute convergence occurs when the series of absolute values of the terms converges.

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7. What happens if an alternating series does not satisfy the conditions for convergence?

A. The series always converges conditionally.

B. The series converges absolutely.

C. The series diverges.

D. The series converges only for even terms.

Answer: C. The series diverges.

Explanation: If an alternating series does not meet the criteria for convergence (decreasing terms approaching zero), it will diverge.

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8. In which case will an alternating series not converge?

A. If the terms tend to zero.

B. If the terms decrease in absolute value.

C. If the terms oscillate but do not tend to zero.

D. If the series has only even terms.

Answer: C. If the terms oscillate but do not tend to zero.

Explanation: An alternating series will not converge if the terms do not tend to zero, regardless of whether they oscillate.

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9. Which of the following is an example of a series that converges conditionally but not absolutely?

A. $\sum \frac{1}{n^2}$

B. $\sum (-1)^n \frac{1}{n}$

C. $\sum \frac{1}{n}$

D. $\sum (-1)^n \frac{1}{n^2}$

Answer: B. $\sum (-1)^n \frac{1}{n}$

Explanation: The series $\sum (-1)^n \frac{1}{n}$ converges conditionally, as the alternating series test holds, but does not converge absolutely because $\sum \frac{1}{n}$ diverges.

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10. What is the sum of an alternating series that satisfies the conditions of the Alternating Series Test?

A. It is always zero.

B. It is the sum of the positive terms.

C. It is the sum of the negative terms.

D. It is a finite value determined by the behavior of the terms.

Answer: D. It is a finite value determined by the behavior of the terms.

Explanation: If the series converges, the sum will be a finite value depending on how the terms behave and tend toward zero.

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1. What is the general form of a Taylor series?

A. $f(x) = \sum_{n=0}^{\infty} f(n)(x-c)^n$

B. $f(x) = \sum_{n=0}^{\infty} \frac{f(n)}{n!}(x-c)^n$

C. $f(x) = \sum_{n=0}^{\infty} \frac{f(c)}{n!}(x-c)^n$

D. $f(x) = \sum_{n=0}^{\infty} \frac{f(x)}{n!}(x-c)^n$

Answer: B. $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!}(x-c)^n$

Explanation: The Taylor series expresses a function as an infinite sum of terms involving derivatives of the function evaluated at a point c.

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2. What is the Maclaurin series?

A. A Taylor series centered at $x = 0$

B. A Taylor series centered at any point c

C. A series that only uses even powers of x

D. A series that approximates a polynomial function

Answer: A. A Taylor series centered at $x = 0$

Explanation: The Maclaurin series is a special case of the Taylor series where the expansion is centered at $x = 0$.

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3. Which of the following is the first term of the Taylor series expansion of e^x around $x = 0$?

A. 1

B. x

C. $\frac{x}{1!}$

D. $\frac{x^2}{2!}$

Answer: A. 1

Explanation: The Taylor series for e^x is $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$, and the first term is 1 (when $n = 0$).

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4. What is the Maclaurin series for $\sin(x)$?

A. $\sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}$

B. $\sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}$

C. $\sum_{n=0}^{\infty} \frac{x^{n}}{n!}$

D. $\sum_{n=0}^{\infty} \frac{x^{n}}{n^2!}$

Answer: A. $\sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}$

Explanation: The Maclaurin series for $\sin(x)$ is $\sin(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$, but the absolute value of the terms matches the answer A.

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6. What is the Maclaurin series for $\cos(x)$?

A. $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$

B. $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$

C. $\sum_{n=0}^{\infty} \frac{x^n}{n!}$

D. $\sum_{n=0}^{\infty} \frac{x^n}{(n+1)!}$

Answer: A. $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$

Explanation: The Maclaurin series for $\cos(x)$ is $\cos(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$.

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Which of the following is the probability of an event that is certain to happen?

- **A) 0**
- **B) 1**
- **C) 0.5**
- **D) -1**

Answer: B) 1

Explanation: A probability of 1 means the event is certain to occur, so the probability is 100%.

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If the probability of event A is P(A)=0.3P(A) = 0.3P(A)=0.3, what is the probability of the complementary event A′A'A′?

- A) P(A′)=1.3P(A') = 1.3P(A′)=1.3
- B) P(A′)=0.7P(A') = 0.7P(A′)=0.7
- C) P(A′)=0.3P(A') = 0.3P(A′)=0.3
- D) P(A′)=1.0P(A') = 1.0P(A′)=1.0

Answer: B) P(A′)=0.7P(A') = 0.7P(A′)=0.7

Explanation: The probability of the complementary event A′A'A′ is given by P(A′)=1−P(A)P(A') = 1 - P(A)P(A′)=1−P(A), so P(A′)=1−0.3=0.7P(A') = 1 - 0.3 = 0.7P(A′)=1−0.3=0.7.

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What does it mean if two events are mutually exclusive?

- **A) They can both happen at the same time.**
- **B) They cannot happen at the same time.**
- **C) One event happens more frequently than the other.**
- **D) Both events are equally likely to occur.**

Answer: B) They cannot happen at the same time.

Explanation: Two events are mutually exclusive if the occurrence of one event prevents the occurrence of the other event at the same time.

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What is the probability of an event if the event is equally likely to occur or not occur?

- **A) 0**
- **B) 1**
- **C) 0.5**
- **D) 0.25**

Answer: C) 0.5

Explanation: When an event is equally likely to happen or not happen, its probability is 0.5, meaning there is a 50% chance of occurrence.

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What is the probability of rolling a 6 on a fair die?

- A) 12\frac{1}{2}21
- B) 16\frac{1}{6}61
- C) 15\frac{1}{5}51
- D) 13\frac{1}{3}31

Answer: B) 16\frac{1}{6}61

Explanation: A fair die has 6 sides, and only one side shows a 6. Therefore, the probability of rolling a 6 is 16\frac{1}{6}61.

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In probability theory, what does the term "sample space" refer to?

- A) The set of all possible outcomes of an experiment.
- B) The set of favorable outcomes for an event.
- C) The probability of an event.
- D) The number of successful events.

Answer: A) The set of all possible outcomes of an experiment.

Explanation: The sample space is the set of all possible outcomes for a particular random experiment.

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SUBJECTIVE PART

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Lecture 23-24: Exercise

Find the absolute maximum and minimum values of given functions on the given closed intervals and state where those values occur.

1) $f(x) = 4x^2 - 4x + 1; \quad [0,1]$
2) $f(x) = 2x^3 - 3x^2 - 12x; \quad [-2,3]$
3) $f(x) = \dfrac{3x}{\sqrt{4x^2+1}}; \quad [-1,1]$
4) $f(x) = 8x - x^2; \quad [0,6]$

Find the absolute maximum and minimum values of the given functions, if any, on the given intervals and state where those values occur.

1) $f(x) = x^2 - 3x - 1 \quad (-\infty, \infty)$
2) $f(x) = 3 - 4x - 2x^2 \quad (-\infty, \infty)$
3) $f(x) = x^3 - 3x - 2 \quad (-\infty, \infty)$
4) $f(x) = \dfrac{x^2}{x+1}; \quad (-5,-1)$

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Practice Questions for Lecture # 25-26

Question # 1. Evaluate the integral $\int \dfrac{\sin 2x}{\cos x}\,dx$

Solution:

$$\int \frac{Sin2x}{Cosx}\,dx = -Cosx + c$$

Question # 2

Evaluate the integral $\int [4\sec^2 x + \csc x \cot x]\,dx$

Solution:

$\int[4\sec^2 x+\csc x\cot x]dx = 4\tan x-\csc x+C$

Question # 3

Evaluate the integral $\int[\frac{1}{t^2}-\cos t]dt$

Solution:

$\int[\frac{1}{t^2}-\cos t]dt = -\frac{t\sin(t)+1}{t}+c$

Question # 4

Evaluate the integral $\int[1+\sin^2\theta\csc\theta]d\theta$

Solution:

$\int[1+\sin^2\theta\csc\theta]d\theta = \theta-\cos(\theta)+c$

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Question # 5

Evaluate the integral $\int[t+\frac{2}{\sin^2 t}]dt$

Solution:

$\int[t+\frac{2}{\sin^2 t}]dt = \frac{1}{2}(t^2-4\cot(t))+c$

Question # 6

Find a function f such that $f''(x)=x+\cos x$ and such that $f(0)=1 and\ f'(0)=2$

Chapter # 24

Q1) If the roots lies between 1 and 2, then find the approximate solution of $x^2 - 2$, by using Newton's method.

Solution:

$$x^4 + x - 3 = 0$$

$$Let\ f(x) = x^4 + x - 3$$

$$Since, \text{f}(1) = -1\ and f(2) = 15, \text{so the root lies between}$$

$$1 \text{and} 2.$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$= x_n - \frac{x_n{}^4 + x_n - 3}{4x^3{}_n + 1} = \frac{3x^4{}_n + 3}{4x^3{}_n + 1}$$

$$f(x) = 1 - x^{\frac{2}{3}}$$

Since our approximations for x_4 and x_5 are equal, all future values will be the same. So the root will be x = 1.414213.

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Q2) Use Newton's method to find the positive solutions of $x^4 + x - 3 = 0$

Solution:

$Since, \mathrm{f}(1) = -1 and f(2) = 15, so the root lies between$

1and 2.

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$x_1 = \frac{1+2}{2} = 1.5$$

$$x_2 = \frac{3(1.5)^4 + 3}{4(1.5)^3 + 1} \approx 1.1654196$$

$$x_3 = \frac{3(1.1654196)^4 + 4}{4(1.1654196)^3 + 1} \approx 1.164037$$

$$x_4 = \frac{3(1.164037)^4 + 4}{4(1.164037)^3 + 1} \approx 1.164035$$

$$x_5 = \frac{3(1.164035)^4 + 4}{4(1.164035)^3 + 1} \approx 1.164035$$

Since our approximations for x_4 and x_5 are equal, all future values will be the same. So the root will be x = 1.164035.

Q3) If f(x) = x(x-1)(x-2), a = 0, b = 0.5, find c of the Mean value Theorem.

Q4) If $f(x) = 3x^2 - 5x + 1$, a = 2, b = 5, find c of the Mean value Theorem.

Q5) Verify Rolle's Theorem for $f(x) = 1 - x^{\frac{2}{3}}$ on [-1,1].

Question NO.01:

Find the arc length of the graph $y = x^{\frac{3}{2}}$ from $(1,1)$ to $(4,8)$.

Hint:

Use the following formula to find the arc length

$$\int_1^4 \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$$

Question NO.02:

Determine the surface area of the solid obtained by rotating $y = \sqrt{9 - x^2}, -2 \le x \le 2,$ about x-axis.

Hint:

Use the following formula to find the surface area

$$\int_{-2}^{2} 2\pi y \sqrt{1+\left(\frac{dy}{dx}\right)^2}\, dx$$

Question NO.03:

Find the work done on a spring when you compress it from its natural length $1m$ to a length of $0.75m$ if the spring constant is $k = 16N \setminus m.$

Hint:

Use the following formula to calculate the work done

$$W = \int_{a}^{b} F(x)\, dx$$

Question NO.04:

Evaluate the following improper integral

$$\int_{-\infty}^{\infty} xe^{-x^2}\, dx$$

Hint:

In this case we've got infinities in both limits and so we'll need to split the integral up into two separate integrals. We can split the integral up at any point, so let's choose $a = 0$ since this will be a convenient point for the evaluation process. The integral is then

$$\int_{-\infty}^{\infty} xe^{-x^2}\, dx = \int_{-\infty}^{0} xe^{-x^2}\, dx + \int_{0}^{\infty} xe^{-x^2}\, dx$$

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Question NO.05:

Evaluate the following limit

$(a)\ \lim_{x\to 0^+} x\ln x$

$(b)\ \lim_{x\to -\infty} xe^{x}$

Hint:

Use L'Hospital's to evaluate the above limits.

Question NO.06

Determine if the following sequence converge or diverge. If the sequence converges determine its limit.

$$\left\{\frac{3n^2-1}{10n+5n^2}\right\}_{n=3}^{n=\infty}$$

Hint:

To do a limit in this form all we need to do is factor from the numerator and denominator the largest power of *n*, cancel and then take the limit as explained below

$$\lim_{n\to\infty}\frac{3n^2-1}{10n+5n^2}=\lim_{n\to\infty}\frac{n^2\left(3-\frac{1}{n^2}\right)}{n^2\left(\frac{10}{n}+5\right)}=\lim_{n\to\infty}\frac{3-\frac{1}{n^2}}{\frac{10}{n}+5}=\frac{3}{5}$$

Question NO.07:

Determine if the following series is convergent or divergent.

$$\sum_{n=2}^{\infty}\frac{1}{n\ln n}$$

Hint:

Apply integral test here to check the convergence or divergence

Question NO.08

Determine if the following series is convergent or divergent.

$$\sum_{n=0}^{\infty}\frac{4n^2-n^3}{10+2n^3}$$

Hint:

Apply divergence test here to check the convergence or divergence

Question NO.09

Determine if the following series is convergent or divergent.

$$\sum_{n=1}^{\infty}\frac{(-10)^n}{4^{2n+1}(n+1)}$$

Hint:

Apply ratio test here to check the convergence or divergence.

Question NO.10

Determine the radius of convergence and interval of convergence for the following power series.

$$\sum_{n=1}^{\infty}\frac{(-1)^n n}{4^n}(x+3)^n$$

Hint:

Apply the ratio test to determine the radius of convergence and interval of convergence.

Question NO.11

Find the Taylor Series for $f(x)=e^{-x}$ about $x=-4$.

Question no 1

Find $f(x_k^*)$ for $f(x) = x$ by dividing the interval [0, 2] into n equal-length subintervals and taking x_k^* as right endpoint of kth subinterval.

Solution:

$$\Delta x=\frac{2-0}{n}=\frac{2}{n}$$

$$\therefore\ x_k^*=a+k\Delta x=0+\frac{2k}{n} \qquad \left(\because\ x_k=a+k\Delta x;\quad for\quad k=0,1,2,...,n \right.$$
$$\left. right\ endpoints:\quad x_k^*=x_k=a+k\Delta x\right)$$

$$f(x_k^*)=x_k^*=\frac{2k}{n}$$

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Question no 2

Find $f(x_k^*)$ for $f(x) = x$ by dividing the interval [0, 3] into n equal-length subintervals and taking x_k^* as left endpoint of kth subinterval.

Solution:

$$\Delta x=\frac{3-0}{n}=\frac{3}{n}$$

$$\therefore\ x_k*=x_{k-1}=a+(k-1)\Delta x=0+\frac{3(k-1)}{n} \qquad \left(\because\ x_{k-1}=a+(k-1)\Delta x;\quad for\quad k=0,1,2,...,n\right.$$
$$\left. left\ endpoint:\quad x_k*=x_{k-1}=a+(k-1)\Delta x\right)$$

$$f(x_k*)=x_k*=\frac{3(k-1)}{n}$$

Question no 3

Find the approximate area under the curve $y = x^3$ over the interval [0,1] by taking

$\Delta x = \frac{1}{n}$ and $x_k^* = \frac{k}{n}$

Solution:

The kth rectangle had area

$$f(x_k^*)\Delta x = \left(x_k^*\right)^3 (\Delta x)$$
$$= \left(\frac{k^3}{n^3}\right)\frac{1}{n}$$
$$= \frac{k^3}{n^4}$$

And the sum of n rectangles is

$$Approximate\ area = \sum_{k=1}^{n} f(x_k^*)\Delta x$$
$$= \sum_{k=1}^{n} \frac{k^3}{n^4}$$
$$= \frac{1}{n^4}\sum_{k=1}^{n} k^3$$
$$= \frac{1}{n^4}\left(\frac{n(n+1)}{2}\right)^2$$

Question no 4

Suppose that the function *h* is integrable such that $\int_{-1}^{1} h(r)\ dr = 3$ and $\int_{4}^{1} h(r)\ dr = -8$. Find $\int_{-1}^{4} h(r)\ dr$

Solution:

$$\int_{-1}^{4} h(r)\ dr = \int_{-1}^{1} h(r)dr + \int_{1}^{4} h(r)dr$$
$$= \int_{-1}^{1} h(r)dr + (-\int_{4}^{1} h(r)dr)$$
$$= 3 + (-(-8)) = 11$$

Question no 5

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Express the limit $\lim_{\max \Delta x_k \to 0} \sum_{k=1}^{n} \left(Cos^2 x_k^* \right) \Delta x_k$ as a definite integral on the interval $[0, \frac{\pi}{2}]$. (Do not evaluate the integral).

Solution:

$$\lim_{\max \Delta x_k \to 0} \sum_{k=1}^{n} \left(Cos^2 x_k^* \right) \Delta x_k = \int_0^{\frac{\pi}{2}} Cos^2 x dx$$

Question no 6

Express the area of the region below the line 6x-7y = -13, above x-axis and between the lines x = 1, x = 4 as a definite integral. Also express this integral as a limit of the Riemann Sum.

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Solution:

$6x - 7y = -13$

$-7y = -13 - 6x$

$7y = 13 + 6x$

$y = \frac{6x + 13}{7}$

$\int_1^4 \frac{6x + 13}{7} dx$

$\int_1^4 \frac{6x + 13}{7} dx = \lim_{\max \Delta x_k \to 0} \sum_{k=1}^{n} \left(\frac{6x_k^* + 13}{7} \right) \Delta x_k$

Question no 7

Use the first fundamental theorem of calculus to evaluate the definite integral: $\int_0^{\pi} \tan t\, dt$

Solution:

$f(t) = \tan t$

$\Rightarrow F(t) = \ln \sec t$

$\Rightarrow F'(t) = f(t)$

$$\int_0^{\pi} \tan t\, dt = \left[\int \tan t\ dt\right]_0^{\pi} \qquad \left(\because \textit{by the First Fundamental Theroem}\ \int_a^b f(x)\,dx = \left[\int f(x)\,dx\right]_a^b\right)$$

$= \ln \sec t \,\big|_0^{\pi} = \ln \sec \pi - (\ln \sec 0) = \textit{undefined}$

Question no 8

Find f_{ave} over the interval [1,5] for $f(x) = \frac{1}{x}$

Solution:

$$f_{ave} = \frac{1}{b-a}\int_a^b f(x)dx = \frac{1}{5-1}\int_2^6 \frac{1}{x}dx$$

$$= \frac{1}{4}\left|\ln x\right|_1^5 = \frac{1}{4}(\ln 5 - \ln 1) = \frac{1}{4}\ln 5$$

Question no.9

Evaluate the integral: $\int \frac{2}{\sqrt{x}.(1+\sqrt{x})^3}dx$ by using a proper substitution.

solution:

$let\ u = 1+\sqrt{x} \Rightarrow du = \frac{1}{2\sqrt{x}}dx$

$\Rightarrow 2du = \frac{1}{\sqrt{x}}dx$

$$\int \frac{2}{\sqrt{x}.(1+\sqrt{x})^3}dx = 4\int \frac{1}{u^3}du$$

$$= 4\left|\frac{u^{-3+1}}{-3+1}\right| = \frac{-2}{u^2}$$

$$= \frac{-2}{(1+\sqrt{x})^2}$$

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Question no.10

Use the Substitution method to express the definite integral in terms of the variable 'u' but do not evaluate $\int_0^{\frac{\pi}{2}} e^{\sin t}\cos t\ dt$

Solution:

$let \quad u = \sin t \quad \Rightarrow du = \cos dt$

$t = 0 \Rightarrow u = \sin 0 = 0$

$and \quad t = \frac{\pi}{2} \Rightarrow u = \sin\frac{\pi}{2} = 1$

$as \quad t\ goes\ from\ 0\ to\ \frac{\pi}{2}$

$so \quad u\ goes\ from\ 0\ to\ 1$

$$\int_0^{\frac{\pi}{2}} e^{\sin t}\cos t\ dt = \int_0^1 e^u\ du$$

Question no.11

If $F(x) = \int_{\frac{\pi}{2}}^{x} \sin t\, dt$, find $F'(x)$ by using the Second Fundamental Theorem of Calculus.

Solution:

$$F(x) = \int_{\frac{\pi}{2}}^{x} \sin t\, dt$$

$$F'(x) = f(x) \qquad \left(\because \text{the Second Fundamental Theorem } F(x) = \int_a^x f(t)\,dt \Rightarrow F'(x) = f(x)\right)$$

$$= \sin x$$

Question no.12

Find a definite integral indicating the area bounded by the curves $y = x + 3$ and $y = \frac{1}{2}x^2$ between $x = -1$ and $x = 2$. But do not evaluate it further.

solution:

Here the function will be

$(x+3)-\frac{1}{2}x^2 \qquad as\,(x+3)>\frac{1}{2}x^2 \;\; for \;\; -1\le x\le 2$

and limits of the integration will be

$x=-1,2$

$\int_{-1}^{2}[(x+3)-\frac{1}{2}x^2]dx$

Question no.13

Find the volume of the solid generated when the region between the graphs of

$f(x)=\frac{1}{3}+x^3$ and $g(x)=x^2$ over the interval [0,3] revolved about the x-axis.

Solution:

Since the volume enclosed between two curves $y=f(x)$ *and* $y=g(x)$ revolve about the x-axis so $V=\int_a^b \pi\left(g^2(x)-f^2(x)\right)dx$

Here $f(x)=\frac{1}{3}+x^3$ *and* $g(x)=x^2$ *as* $f(x)\ge g(x)$ *for* $0\le x\le 3$

$a=0$, $b=3$, so

$$V=\int_0^3 \pi\left((\frac{1}{3}+x^3)^2-x^4\right)dx$$

$$V=\int_0^3 \pi(\frac{1}{9}+x^6+\frac{2}{3}x^3-x^4)dx$$

Question no.14

Use cylindrical shells to find the volume of the solid generated when the region 'R' enclosed between $y=2x+1$ and $y=-2x-3$ in the interval [1,3] is revolved about the $y-axis$.

Here the curves are $y = 2x+1$ and $y = -2x-3$

limits of integration are $x = 1$ *and* $x = 3$

$$So\ V = \int_1^3 2\pi x\big((2x+1)-(-2x-3)\big)dx \Rightarrow V = 2\pi\int_1^3 x(2x+1+2x+3)dx$$

$$= 2\pi\int_1^3 x(4x+4)dx \Rightarrow 2\pi\int_1^3 (4x^2+4x)dx$$

$$\Rightarrow 2\pi\left|\frac{4x^3}{3}+\frac{4x^2}{2}\right|_1^3$$

Question no.15

If f is a smooth function on $[1,4]$, then find a definite integral indicating the arc length of the curve $x = y^{\frac{1}{3}}$ from $y = 1$ *to* $y = 4$.
But do not evaluate it further.

solution:

Here $x = g(y) = y^{\frac{1}{3}}$

$\Rightarrow$ $g'(x) = \frac{1}{3}y^{-\frac{2}{3}}$

$\therefore$ $L = \int_1^4 \sqrt{1+\frac{1}{9}y^{\frac{-4}{3}}}\ dy$ $\left(\because \textit{Arc length}\ L = \int_a^b \sqrt{1+[g'(y)]^2}\ dy\right)$

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